HUABANG ZHANG

San Jose, CA • winbondzhang@gmail.com • Portfolio

EDUCATION

University of California, Santa Cruz Santa Cruz, CA
B.S. Robotics Engineering, Minor in Electrical Engineering
Expected Dec 2026 · GPA: 3.81 / 4.0
Relevant Coursework: Mechatronics (ECE 118), Microprocessor System Design (ECE 121), Sensing & Sensor Technologies (ECE 167), Robotics Capstone (ECE 129), Feedback Control (ECE 141), Logic Design (CSE 100), Circuit Analysis (ECE 101)

TECHNICAL SKILLS

Programming Languages: C, C++, Python, MATLAB
Embedded Systems: PIC32MX, STM32F411RE / F446RE (Nucleo), NVIDIA Jetson Orin Nano, ESP32-C3, BASYS3 FPGA
Hardware & Interfaces: UART, SPI, I²C, PWM, ADC/DAC, GPIO, Interrupts, Motor Drivers (H-Bridge), Oscilloscope, Multimeter
Robotics & Controls: Forward/Inverse Kinematics, Hierarchical State Machines, Sensor Calibration & Filtering, Teleoperation
Tools: EasyEDA, Altium Designer, Cadence, Onshape, MPLAB X, PlatformIO, VS Code, Vivado, Git/GitHub, Linux CLI
Fabrication & Testing: PCB Design & Soldering, FDM 3D Printing, Laser Cutting, Custom Test Fixtures, System Validation Testing

ENGINEERING PROJECTS

Mobile Manipulator for Warehouse Assistance (ROS 2) | Independent Project *Jul 2026 – Present*

- Designing a mecanum-wheel mobile manipulator for warehouse pick-and-place as a self-directed follow-up to my capstone: Jetson Orin Nano for perception and ROS 2 planning, STM32 for real-time motor and servo control over UART.
- Modeled and 3D-printed the chassis in Onshape; selected 4× JGB37-520 Hall-encoder gear motors, a 5-DoF arm (5× HX-12H servos) with an HTS-21H gripper, an Aurora 930 Pro depth camera, and a Coin-D6 LiDAR.
- Designed a custom power distribution PCB in EasyEDA for an 11.1 V 3S Li-ion pack (6000 mAh, 3C): motors fed directly from the battery, with buck regulation to 6 V (servos), 9 V (Jetson), and 7 V (STM32).
- In progress: self-teaching ROS 2 and YOLO to bring up base teleoperation and depth-fused object detection; long-term goal is natural-language voice control via an on-device or cloud LLM.

Assistive 7-DoF Robotic Arm for Daily Living | Electrical Lead, Senior Capstone *Sep 2025 – Jun 2026*

- Led the electrical subsystem in a 4-person team building a 7-DoF assistive arm (250 g payload) that helps people with limited hand function cook and fetch items; integrated STM32 servo control, Jetson Orin Nano AprilTag vision, and an ESP32 joystick.
- Architected a multi-rail power system from a 19.5 V / 240 W input: four 7.0 V buck branches sharing the servo load, dedicated 11.5 V (Jetson) and 8.75 V (Nucleo) rails, and fuse/TVS protection.
- Designed custom PCBs (buck converters, power distribution, Nucleo I/O shield, controller BMS) through 3 faculty-reviewed iterations, improving noise immunity with a compact ground-referenced feedback loop and wide copper power pours.
- Validated power integrity by oscilloscope: ripple of 0.53% (Jetson), 0.76% (STM32), and 2.2% on the servo rail under dynamic 250 g load; sustained 10 min of full-speed 6-servo motion with no converter overheating, brownout, or controller reset.
- Wrote low-level joint control in C++ converting kinematic joint-angle targets into calibrated servo commands; reached ~85 ms average command latency (250 ms target) and 10–25 mm accuracy, limited by open-loop servos.

4-DoF Teleoperated Robotic Arm (IMU + Flex Sensors) | Embedded Lead, ECE 167 Final Project *Jan – Mar 2026*

- Built the sensing foundation in C on STM32F411RE: interrupt-driven firmware reading IMU, flex/FSR, encoder, and ultrasonic sensors via I²C, ADC, PWM, and interrupts, achieving 3.75° encoder resolution.
- Calibrated the IMU with bias/scale correction and ellipsoid fitting, and conditioned sensor signals with IIR filtering and regression, reducing drift and noise in the sensor-to-actuator mapping.
- Developed gesture-based teleoperation mapping IMU orientation and flex/FSR readings to PWM servo commands for 1:1 motion mimicry; achieved reliable pick-and-place in <30 s (2× faster than requirement).
- Eliminated servo jitter by refining the C control loop (deadband, low-pass filtering, rate limiting); co-designed dual-buck power distribution with per-servo decoupling to hold voltage under >6 A stall current.

Autonomous Package Sorting Conveyor System | Mechanical Lead, ECE 118 Final Project *Apr – Jun 2025*

- Built an autonomous conveyor that sorts packages into 6 categories (3 colors × heavy/light) using a load cell (HX711), TCS3200 color sensor, IR beam-break sensors, and servo pushers, controlled by hierarchical state machines in C (ES Framework on PIC32).
- Owned all mechanical design in Onshape: belt chassis, load-cell weight station, servo pusher arm, swappable sorting slots, and a tension-adjustable roller mount; built mainly from laser-cut plywood, 3D-printing only rollers, pusher arm, and servo couplers.
- Resolved weight misclassification from inconsistent handling (dropped light packages read as heavy; slowly released heavy packages read as light) by adding a settling delay before load-cell sampling.
- Implemented sensor-gated flow control that weighs and releases the next package only after beam-break sensors confirm the belt is clear, sorting with 100% accuracy at 15–18 packages/min (2.5–3× the 6/min requirement).